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Research note

Palaeomagnetism and ages of dolerite dykes in Karimnagar District, Andhra Pradesh, India

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Summary. Results of palaeomagnetic investigations of three dolerite dykes in Karimnagar District, Andhra Pradesh, India are reported. The remanent magnetic directions of specimens from one dyke, called A, fall into two groups, one with a positive inclination and the other with a negative inclination, indicating two intrusions. For dykes B and C, the remanent magnetic directions show a negative inclination. The pole positions for all these dykes are close to those for the Gwalior Traps, whose age is 1830±200 Myr. It is inferred that these dykes were emplaced over a short span of time around the Gwalior Trap period, which may represent a phase of multiple igneous activity.

Introduction

It is a common practice to determine the age of rocks by correlating their palaeomagnetic pole positions with the data for formations of known ages. Palaeomagnetic investigations were carried out over three dolerite dykes in Karimnagar district, Andhra Pradesh, India, mainly to substantiate the interpretation of magnetic anomalies over them. The data thus obtained are, however, used to ascertain the palaeomagnetic age of these dykes.

Geology, location and sampling

The geological map of the area, situated between latitudes 18°20′ and 18°30′N and longitudes 79°0′ and 79°15′E, is shown in Fig. 1. The area is mostly occupied by grey granitic gneisses and granites which grade into one another. There are three major dolerite dykes, named A, B and C (Fig. 1), intruding into the granitic rocks. Magnetic surveys conducted in this area indicate that they run for several kilometres in a N60°E direction (Subba Rao 1974), although they appear as disconnected ridges on the surface. They are 35–100 m wide and have a vertical dip. The age of the dolerite dykes has not been determined by any method. Crawford (1969), however, believes that they are younger than the granites (2000 Myr) into which they intrude.

As the primary objective of the palaeomagnetic study was to substantiate the interpretation of magnetic anomalies and also due to lack of suitable exposures, 24 samples were collected from three sites over dykes A and C, and two sites over dyke B. In all, 48 specimens were prepared and

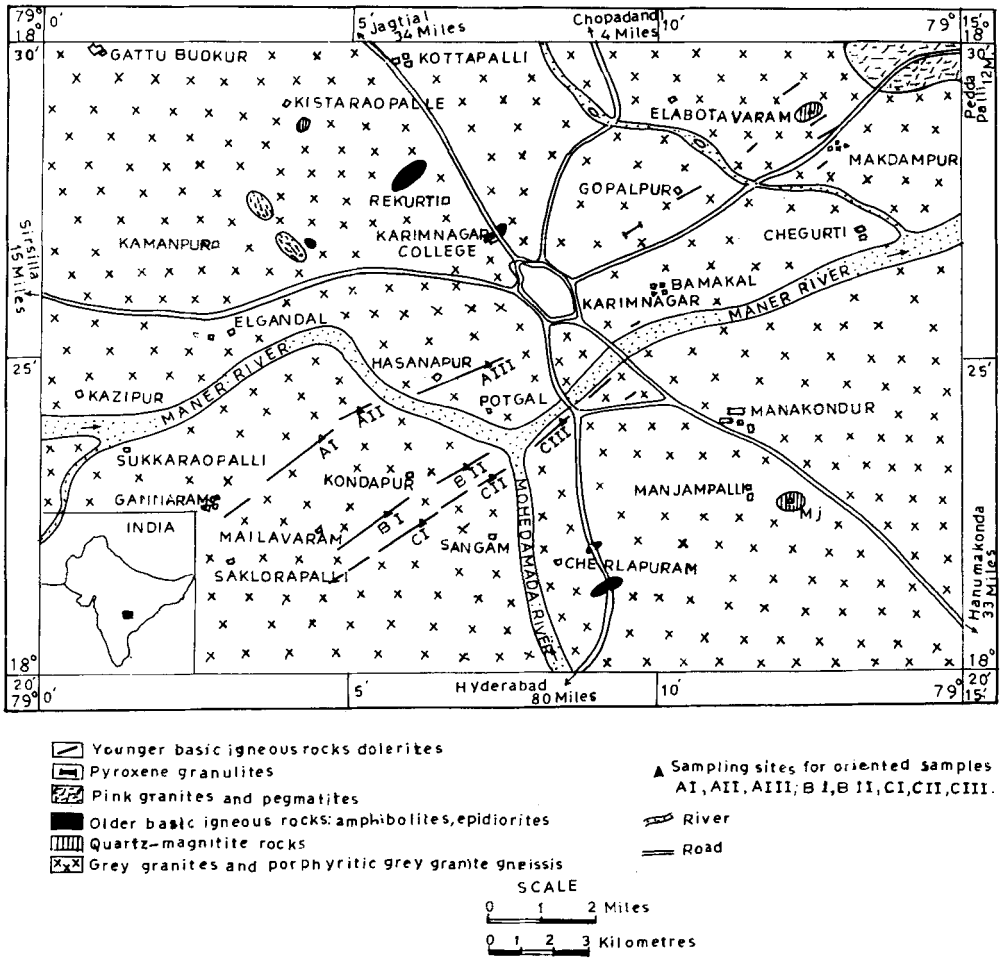


Figure 1. Geological map of the area showing the dykes and distribution of sampling sites.

subjected to palaeomagnetic tests in the laboratory. Each specimen was given a code number, such as AIII 2(1), identifying the dyke, sampling locality, sample number and the specimen number prepared from the same sample.

NRM and demagnetization studies

The magnitude and direction of natural remanent magnetization (NRM) were determined on an astatic magnetometer. The directions of NRM are all highly scattered (Fig. 2), indicating the presence of secondary components of magnetization. For the alternating field (af) demagnetization test, therefore, 12 specimens were initially subjected to progressively increasing alternating magnetic fields of 2.5, 5, 10, 20, 40 and 60 mτ. The demagnetization characteristics for these specimens (Fig. 3) show that they lost most of their NRM intensity in fields less than 40 mτ, beyond which there is no loss in intensity. The magnetic vectors vary randomly in fields less than 40 mτ, remain constant afterwards and show stable end points. Therefore, all the remaining specimens were subjected to the af demagnetization test at 40 mτ (Fig. 2).

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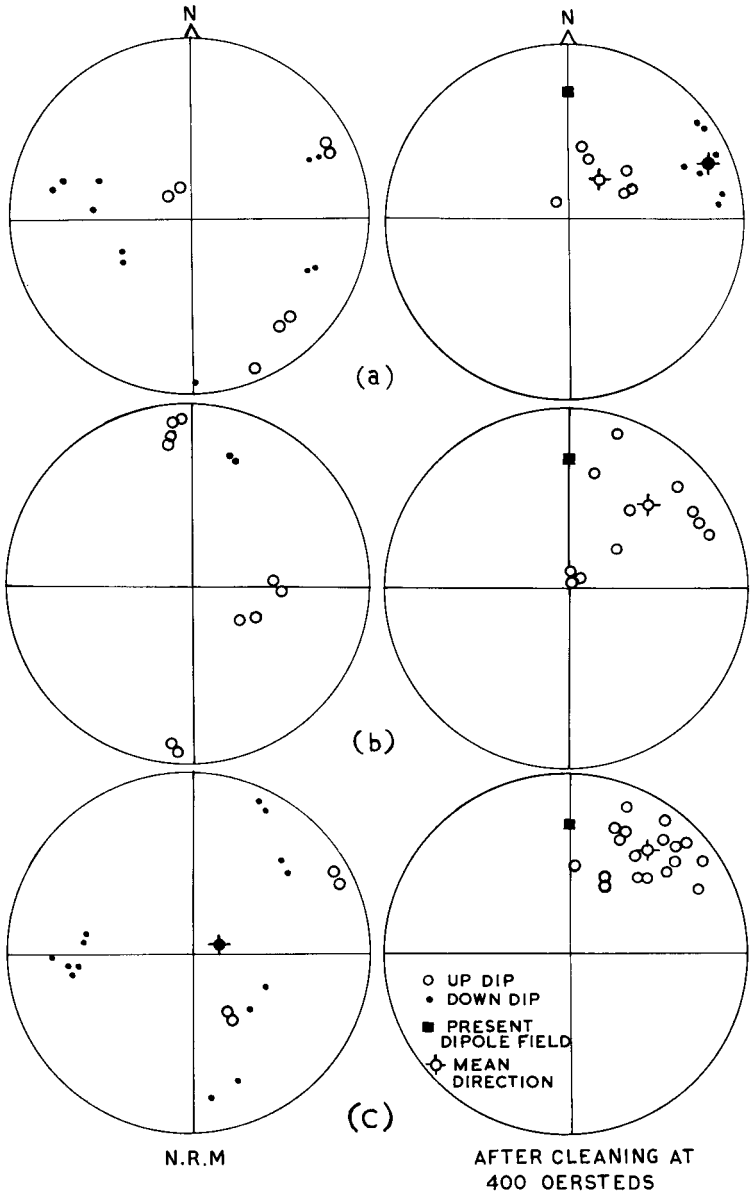


Figure 2. Equal area projections of NRM and stable vectors of specimens from the three dykes A, B and C. The stable vectors shown are obtained after af demagnetization at 40 mT peak field. Open (closed) circles are −ve (+ve) inclinations. The mean of the vectors and present-day dipole field at the site are also shown.

Results and discussion

For dykes B and C, the cleaned directions have a negative inclination (Table 1). For dyke A, the cleaned directions fall into two groups, one with a positive inclination and the other with a negative inclination. Actually, the magnetic anomalies across this dyke had to be explained by two juxtaposing dyke models, one with a positive inclination of magnetization and the other with

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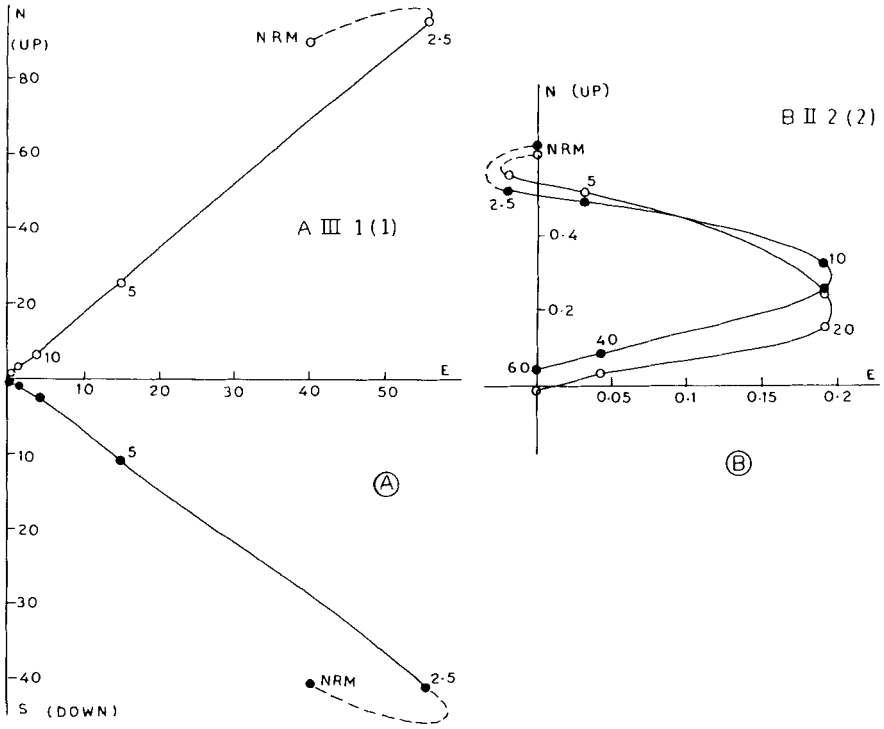


Figure 3. AF demagnetization behaviour of two typical pilot specimens studied from the Karimnagar dykes. Solid and open circles show the vertical and horizontal projections of the remanent vectors. Numbers indicate the peak alternating field in mT. Numbers along the axes refer to magnetization intensity in $A\,m^{-1}$.

a negative inclination of magnetization (Subba Rao & Radhakrishna Murthy 1980). Thus, based on the palaeomagnetic and magnetic interpretations, it may be concluded that dyke A, although appearing as a single outcrop, consists of two individual intrusions with different directions of magnetization.

On the basis of geologic, field, petrographic and palaeomagnetic evidence, four to five phases of dyke emplacement have been suggested by Pascoe (1965), Anjanappa (1975) and others in the Indian Peninsula over a wide span of time from the closing stages of the Archaean until the Cuddapah period. To ascertain the phase of emplacement of these dykes, the present results are compared with the data of those formations for which information on age is available (Table 2). The remanent magnetic directions and VGP positions for the two portions of the dyke A having positive and negative inclinations are in approximate agreement with those of the Gwalior traps. These traps were dated at 1830±200 Myr by Crawford & Compston (1970). The two portions of dyke A could therefore be dated at 1830±200 Myr and they might have been emplaced in two different phases during which period the geomagnetic field was different.

Similarly, the remanent directions and VGP positions of dykes B and C are in general agreement with those of the Kondapalli charnockites, Kolar Dykes and the Hyderabad Dyke (Table 2). Aswathanarayana (1964) showed by radiometric dating techniques that the Kondapalli charnockites were emplaced around 1300–1520 Myr and that their metamorphism and uplift occurred around 500 Myr. Similarly, the Kolar dyke from the Nundydroog mines (Reddy & Prasad 1979) was considered to represent either Precambrian or Deccan trap activity. The Deccan trap activity suggestion was, however, ruled out as the remanent magnetic directions and

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Table 1. Results of palaeomagnetic measurements on dykes from the Karimnagar area.

Dyke	No. of samples N	No. of specimens n	Site coordinates		Remanent magnetic directions		R	K	α_{95}	Palaeomagnetic pole		δ_p	δ_m
			Lat. (°N)	Long. (°E)	D	I				λ_p	L_p		
A (positive group)	9	7	18.4	79.1	67.5	11.7	6.8	35.3	8.9	23.2N	169.9E	4.6	9.0
A (negative group)		6	18.4	79.1	40.1	−62.9	5.8	21.7	12.3	17.3N	129.7W	15.3	19.3
B	6	11	18.4	79.1	45.3	−41.7	9.2	5.4	18.2	28.8N	148.7W	13.6	22.3
C	9	18	18.4	79.1	39.6	−24.6	16.9	15.0	8.5	40N	155.1W	4.9	9.1

D=mean declination east of north.
I=mean inclination, positive downwards.
R=resultant of n unit vectors.
K=precision parameter.
α_{95}=radius of the circle of confidence at 95 per cent probability.
λ_p, L_p=palaeocoordinates of the pole.
δ_p, δ_m=semi-axes of oval of confidence.

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Table 2. Palaeomagnetic data for some Precambrian formation from peninsular India.

Sl. no.	Formation	Lat. (°N)	Long. (°E)	D	I	α_{95}	λ_p	L_p	δ_p	δ_m	Reference
1	Gwalior Traps (1830±200 mA)*	26.0	78.0	70	+3	18.0	19N	176E	5.0	10.0	Athavale, Radhakrishna Murthy & Sehasrabudhe (1963)
2	Gwalior Traps (1830±200 mA)*	26.0	78.0	79	+35	5.0	19N	156E	3.0	6.0	Klootwijk (1974)
3	Gwalior Traps (1830±200 mA)*	26.0	78.0	80	+8	13.0	11N	169E	6.0	13.0	Klootwijk (1974)
4	Gwalior Traps (1830±200 mA)*	26.0	78.0	3	−74	6.5	5N	103W	10.5	11.7	Klootwijk (1974)
5	Gwalior Traps (1830±200 mA)*	26.0	78.0	80	+18	4.0	13N	164E	2.0	4.0	McElhinny, Cowley & Edwards (1978)
6	Gwalior Traps (1830±200 mA)*	26.0	78.0	79	+26	6.0	16N	161E	4.0	7.0	McElhinny, Cowley & Edwards (1978)
7	Hyderabad Dyke	17.4	78.5	44	−3	5.0	43N	173W	2.5	5.0	Verma *et al.* (1968)
8	Hyderabad Dyke	17.4	78.5	32	−16	7.0	49N	155W	–	–	Kumar & Rao (1979)
9	Kolar Dyke P	13.0	78.0	37	−18	2.0	47N	163W	–	–	Reddy & Prasad (1979)
10	Kondapalli-charnockite (1300–1520 mA)†	17.5	83.0	45	−20	–	38N	158W	–	–	Bhimasankaram (1964)

Symbols are as in Table 1.
*Crawford & Compston (1970). †Aswathanarayana (1964).

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VGP positions of the Kolar dyke are different from those of the Deccan traps. The Hyderabad dyke was considered to be contemporaneous with the Gwalior Traps (Verma, Pullaiah & Hasnain 1968; Kumar & Rao 1979). Hence, it is possible that dykes B and C might also have been emplaced during this period. Then the three dykes may represent a phase of multiple igneous activity over a small interval of time, around the Gwalior Trap period, during which there was a change in the Earth's magnetic field direction.

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References

Anjanappa, K., 1975. Palaeomagnetism and age of the dolerite dykes of the Tirupathi area, Chittor District Andhra Pradesh, *Recent Res. Geol.*, **2,** 162.

Aswathanarayana, U., 1964. Isotopic ages from the Eastern Ghats and Cuddapahs of India, *J. geophys. Res.*, **69,** 3479.

Athavale, R. N., Radhakrishna Murthy, C. & Sehasrabudhe, P. W., 1963. Palaeomagnetism of some Indian rocks, *Geophys. J. R. astr. Soc.*, **7,** 304.

Bhimasankaram, V. L. S., 1964. A preliminary investigation on the palaeomagnetic directions of the charnockites of Andhra Pradesh, *Curr. Sci.*, **33,** 465.

Crawford, A. R., 1969. India, Ceylon and Pakistan, new age data and comparisons with India, *Nature,* **223,** 380.

Crawford, A. R. & Compston, W., 1970. The age of the Vindyan system of Peninsular India, *Q. Jl geol. Soc. Lond.,* **125,** 351.

Klootwijk, C. T., 1974. Palaeomagnetic data from the Precambrian Gwalior Traps, Central India, *Tectonophys.*, **21,** 185.

Kumar, A. & Rao, N. T. V. P., 1979. *Proc. 3rd Workshop, Status, Problems and Programmes in Indian Peninsular Shield by IIPG,* p. 146.

McElhinny, M. W., Cowely, J. A. & Edwards, D. J., 1978. Palaeomagnetism of some rocks from Peninsular India and Kashmir, *Tectonophys.,* **50,** 41.

Pascoe, E. H., 1965. *A Manual of the Geology of India and Burma,* Government of India Press, Calcutta 1.

Reddy, V. D. & Prasad, C. V. R. K., 1979. Palaeomagnetism of the dykes from Kolar gold mines, India, *J. geol. Soc. India,* **20,** 489.

Subba Rao, Y. V., 1974. Some geological and geophysical investigations around Karimnagar Town, Andhra Pradesh, (unpublished) *PhD thesis,* Andhra University, Waltair.

Subba Rao, Y. V. & Radhakrishna Murthy, I. V., 1980. Magnetic and palaeomagnetic investigations over a dolerite dyke near Hasanapur Village in Karimnagar District, Andhra Pradesh, *Geophys. Res. Bull.,* **18,** 9.

Verma, R. K., Pullaiah, G. & Hasnain, Iqbal, 1968. Palaeomagnetic study of a dyke near Hyderabad, India, *Bull. natn Geophys. Res. Inst.,* **6,** 79.

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